AI-Powered Diagnostic Assistant for
Diabetic Retinopathy
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Abstract
Diabetic retinopathy is a leading cause of blindness among working-age adults. Early detection through automated screening can prevent vision loss. This project proposes a convolutional neural network pipeline that analyzes optical coherence tomography images to classify retinal conditions into normal, diabetic macular edema, drusen, and choroidal neovascularization categories. We compare the performance of VGG16, ResNet50, and a custom lightweight CNN. The best model achieves 94.7% test accuracy on the OCT2017 dataset. A Flask web interface allows ophthalmologists to upload scans and receive predictions with attention heatmaps and confidence scores.

1. Introduction
Diabetic retinopathy (DR) affects more than 93 million people worldwide. The
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